IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

IBM Data Science Capstone

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04/23/26
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GitHub Link

Outline

- Executive Summary
- Introduction
- Methodology
- Data Collection & Wrangling
- EDA
- Predictive Analysis & Interactive Analytics
- Insights from EDA
- Predictive Modeling
- Conclusions
- Appendix

Executive Summary

- Primary Methodologies
  - Collected SpaceX launch data via API and web scraping
  - Performed data wrangling and feature engineering
  - Conducted EDA with both visualization and SQL queries
  - Built interactive dashboards
  - Developed classification models to predict launch success
- Key Results
  - Launch success rate increases over time
  - Certain launch sites, such as KSC LC-39A showed higher success rates
  - Payload mass, booster version, and orbit type significantly influence success
  - The best classification model was logistical regression and achieved 84.6% accuracy

Introduction

Project background and context

- SpaceX aims to reduce launch costs through reusable rockets. Launch success rates and booster landings are critical for cost efficiency. Understanding the factors that influence the success rate can improve mission planning.

Problem Statement

- What factors most influence SpaceX launch success?
- Can we predict which launches will be successful?
- How do payload, orbit, and launch site impact outcomes?

Section 1

Methodology

Methodology Overview

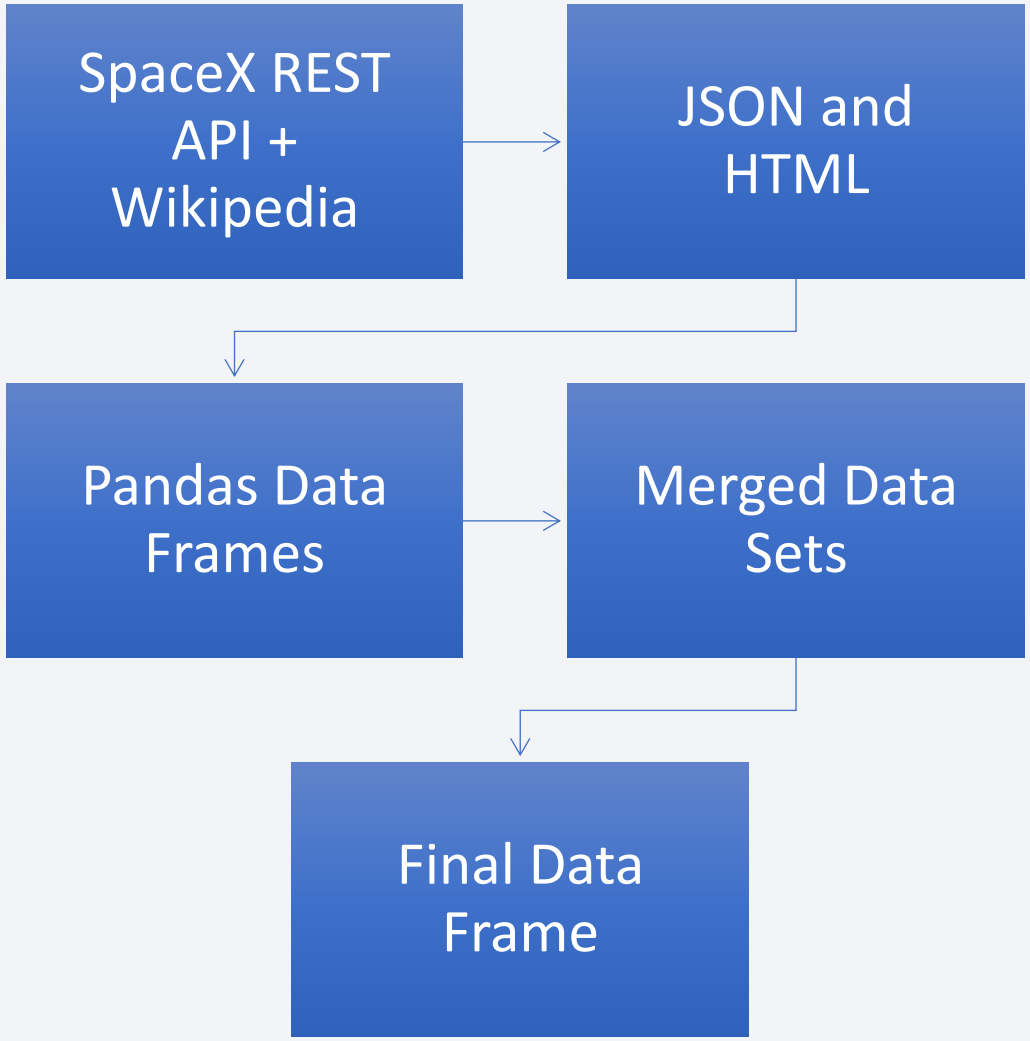
Executive Summary

- Data collection:
  - Collected SpaceX Launch data through REST API
  - Additional data was extracted from Wikipedia using web scraping
- Data Wrangling
  - Cleaned missing and inconsistent values. Created features.
- Exploratory Data Analysis (EDA)
  - Used visualizations (scatter plots, bar charts) and used SQL queries to analyze trends
- Interactive Visual Analytics using Folium and Plotly Dash
  - Built Folium maps to visualize launch locations
  - Developed a Plotly Dash dashboard for dynamic exploration
- Perform predictive analysis using classification models
  - Trained classification models (Logistical Regression, SVM, etc.) and tuned hyperparameters
  - Selected the best model based on Accuracy

Pipeline: Data > Cleaning > EDA > Visualization > Modeling > Insights

Data Collection

- Collected SpaceX launch data from various sources
  - SpaceX REST API
  - Wikipedia
- Retrieved launch details
  - Flight numbers
  - Launch sites
  - Payloads
  - Orbit
  - Outcomes
- Combined data sources into unified dataset

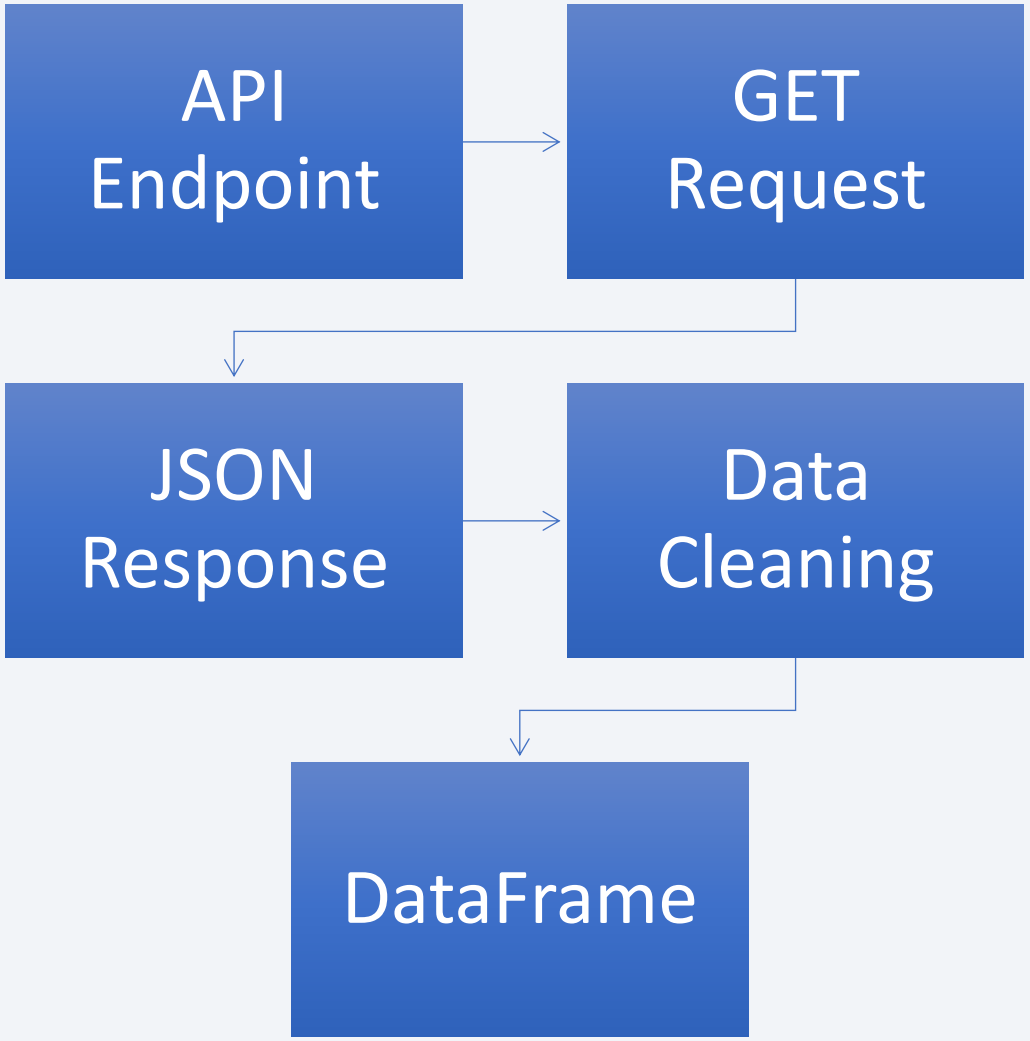


Data Collection – SpaceX API

- Used SpaceX REST API to collect launch data to extract:
  - Launch dates
  - Rocket/Booster versions
  - Payload mass
  - Launch outcomes
- Converted JSON responses into structured data frame

GitHub Link:

- [SpaceX API Data Collection Notebook]

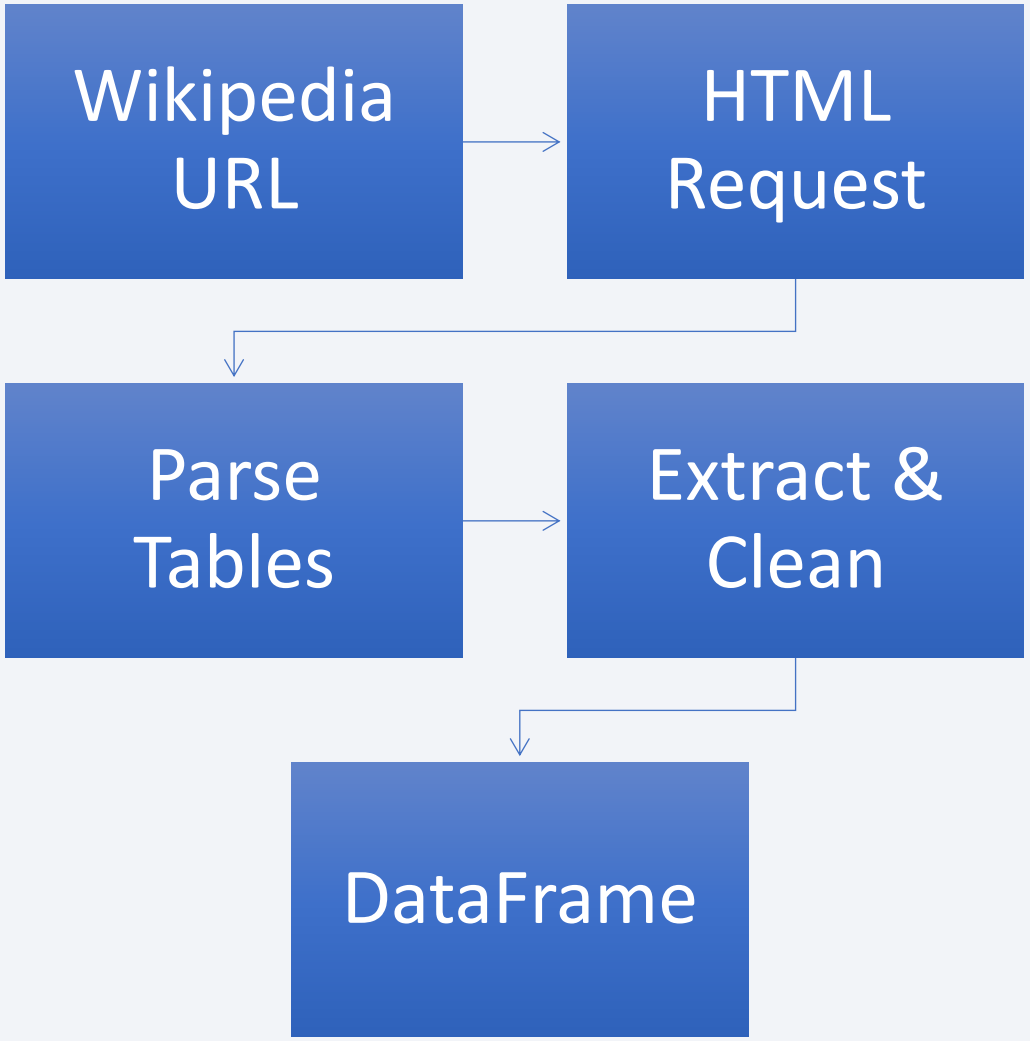


Data Collection - Scraping

- Scraped launch data from Wikipedia
- Used BeutifulSoup to parse HTML files
- Extracted:
  - Launch site names
  - Booster versions
  - Payload and orbit information
- Cleaned and formatted data

GitHub Link:

- [SpaceX Web Scraping Notebook]

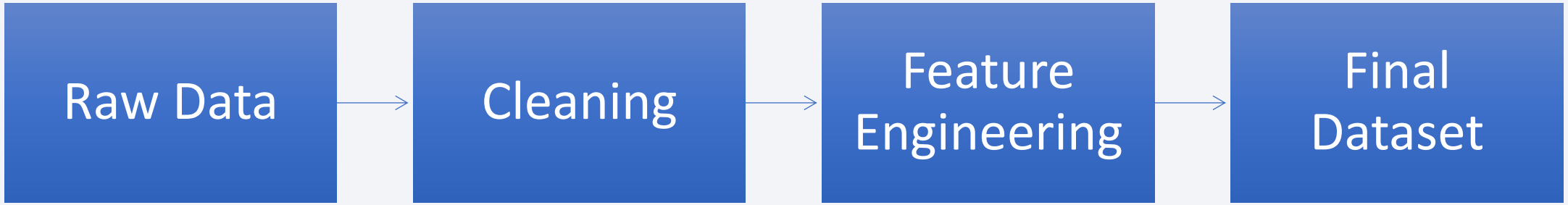


Data Wrangling

- Merged datasets from API and web scraping
- Handled missing values
- Converted data types (e.g. dates, payloads)
- Created target variable (Launch success 1/0)
- Feature engineering

GitHub Link:

- [SpaceX Data Wrangling Notebook]



EDA with Data Visualization

- Charts Used:
  - Scatter Plots:
    - Flight Number by Launch Site
    - Flight Number by Orbit Type
    - Payload Mass by Launch Site
  - Bar Plots:
    - Success Rate by Orbit
  - Line Charts:
    - Success Rate over Time

GitHub Link:

- [EDA with Data Visualization Notebook]

- Why These Charts:
  - Identify relations between variables (payload, orbit, site)
  - Detect trends over time
  - Compare success rate across categories
  - Explore correlations between payload mass and success

EDA with SQL

- Main Queries:
  - Retrieved unique launch site names
  - Filtered sites by name
  - Calculated payload mass by organization (NASA)
  - Computed average payload for specific boosters
  - Identified the first successful landing
  - Listed total number of successes and failures
  - Ranked landing outcomes over time ranges
- Why SQL was used:
  - Efficiency in querying structured data
  - Aggregation and filtration for insights

GitHub Link:

- EDA with SQL Notebook

Build an Interactive Map with Folium

- What was built:
  - Created markers for all SpaceX launch sites
  - Used color-coded markers for success vs failures
  - Drew lines to show proximity to a coastline and city
- Why it was built:
  - Visualize geographic distribution of launch sites
  - Analyze proximity to infrastructure and coastlines
  - Improve spatial understanding of launches and outcomes

GitHub Link:

- Interactive Launch Site Map Notebook

Build a Dashboard with Plotly Dash

- Features:
  - Dropdown menu to select the launch site
  - Interactive pie chart showing launch success rates
  - Scatter plot of payload vs launch outcome
  - Range slider to filter payload mass
- This enabled:
  - Dynamic exploration of data
  - Filtering by launch sites and payload mass
  - Helped identify patterns in success rate
  - Provides a user-friendly interface for analysis

GitHub Link:

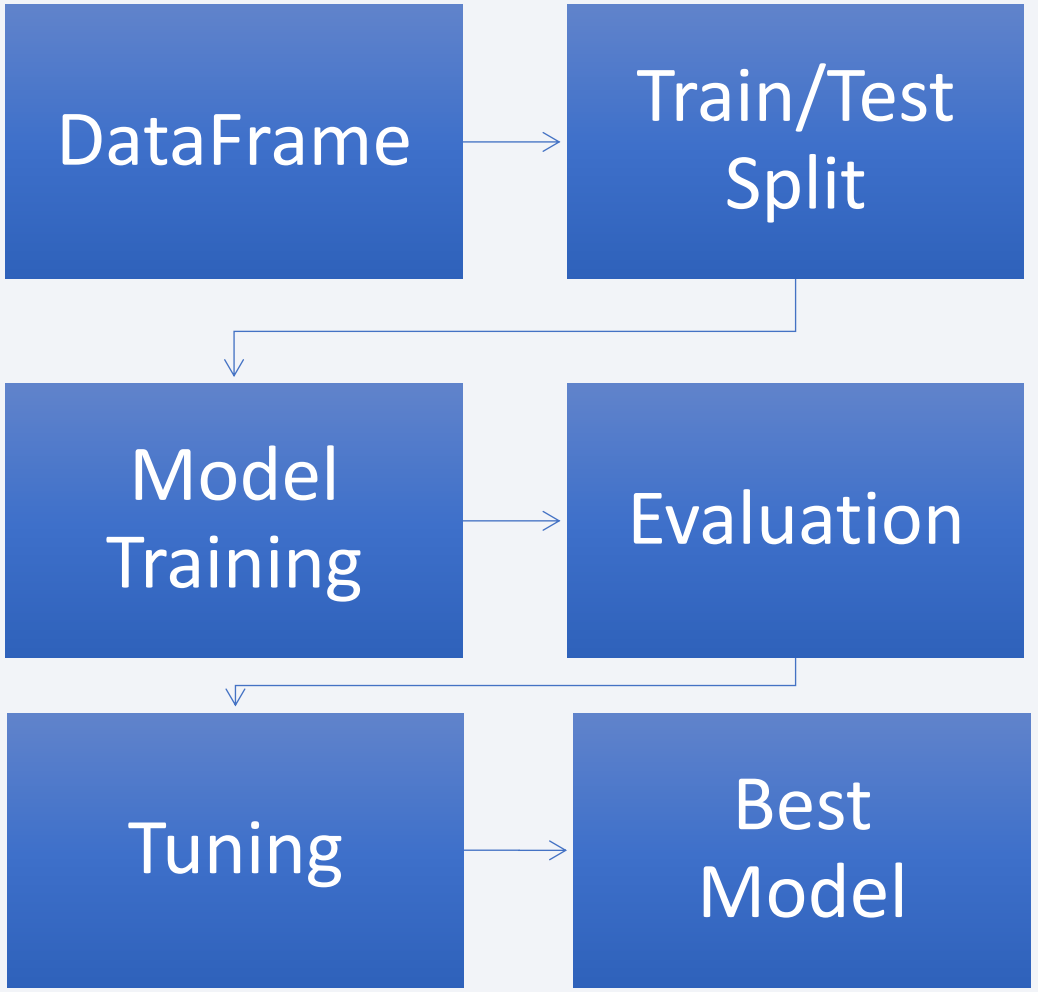
- [Dashboard With Plotly Dash Source Code]

Predictive Analysis (Classification)

- Split data into training and test sets
- Standardized features for modeling
- Trained multiple classification models including:
  - Logistic Regression
  - Support Vector Machine (SVM)
  - Decision Tree
  - K-Nearest Neighbors (KNN)
- Using accuracy as the primary metric
  - Compared performance across models
  - Created a confusion matrix for analysis
- Tuned parameters and selected the best model

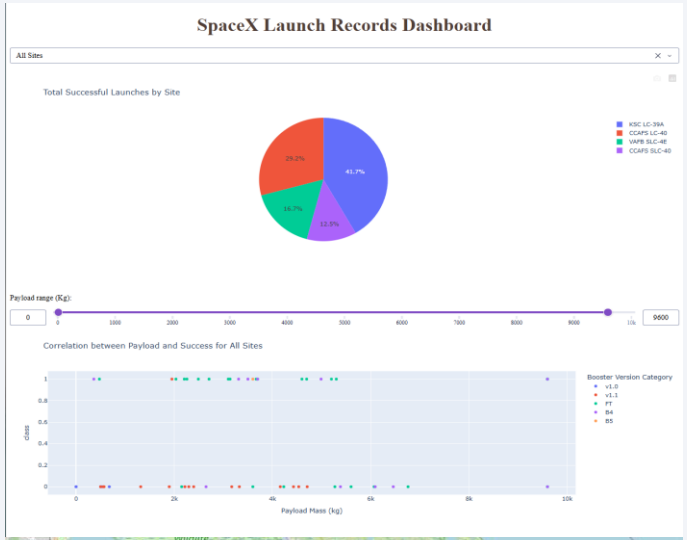
GitHub Link:

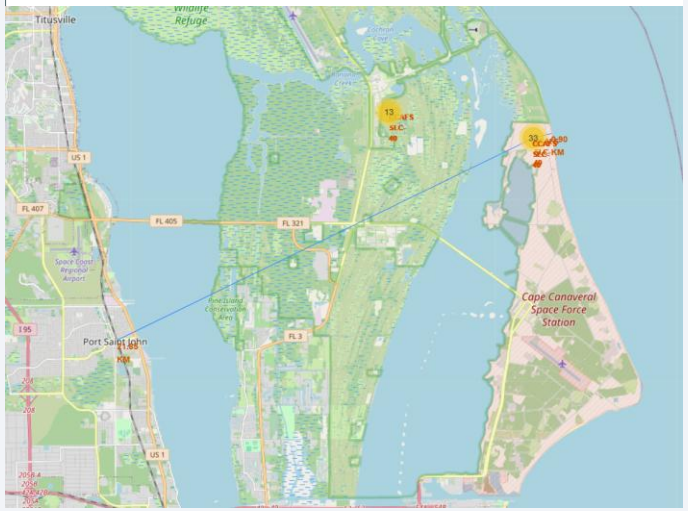
- [ML Predictive Analysis Notebook]



Results

- Exploratory data analysis (EDA) results
  - Identified relationships between payload, orbit, and success
  - Observed increased launch success over time
  - Found variation in success rates across launch sites
- Interactive analytics demo in screenshots
  - Folium map & demo app dashboard included
- Predictive analysis results
  - Multiple classification models trained and evaluated
  - Best model was the decision tree with ~94% accuracy
  - Key Predictors: payload mass, orbit type, booster version

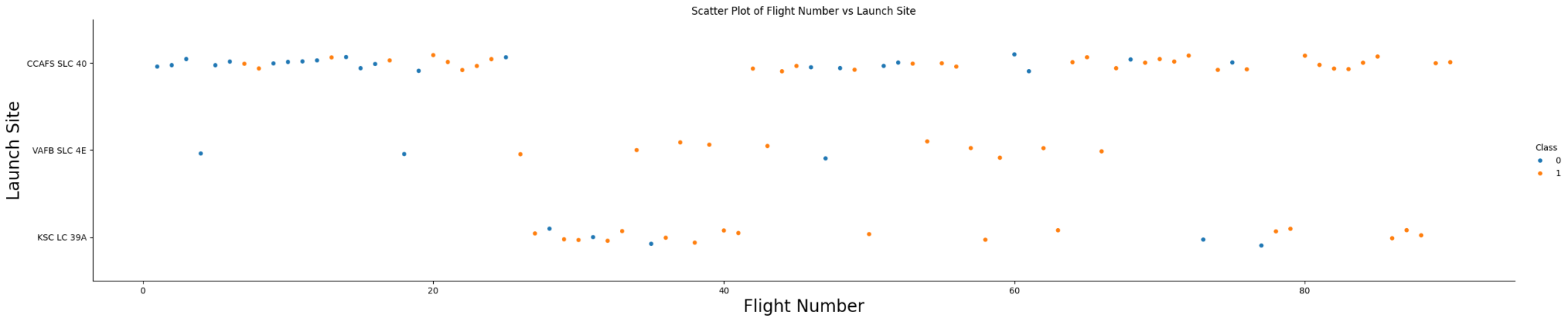




Section 2

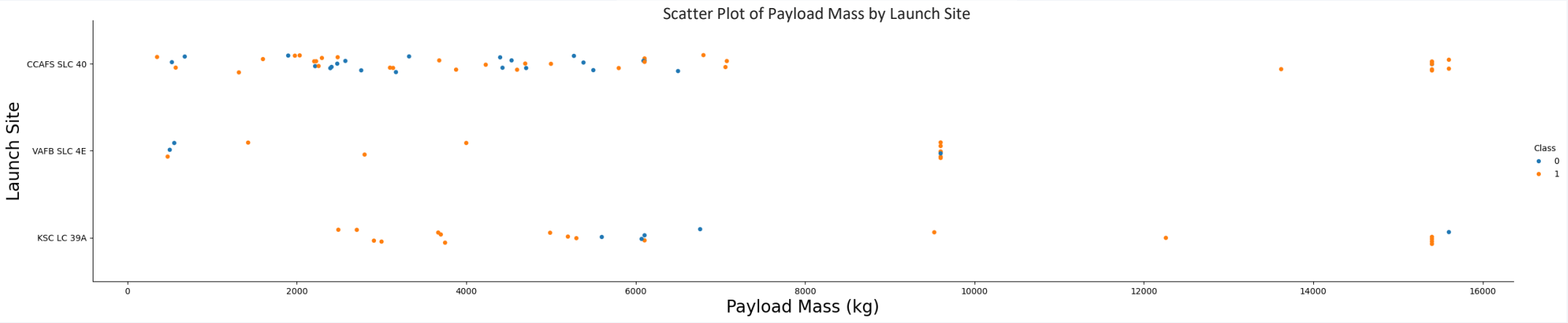
Insights drawn from EDA

Flight Number vs. Launch Site



- This plot shows the flight number by launch site
  - The class is a 0 for failure and a 1 for success
- Insights:
  - Majority of launches come from CCAFS SLC 40
  - Relatively similar success rates across sites

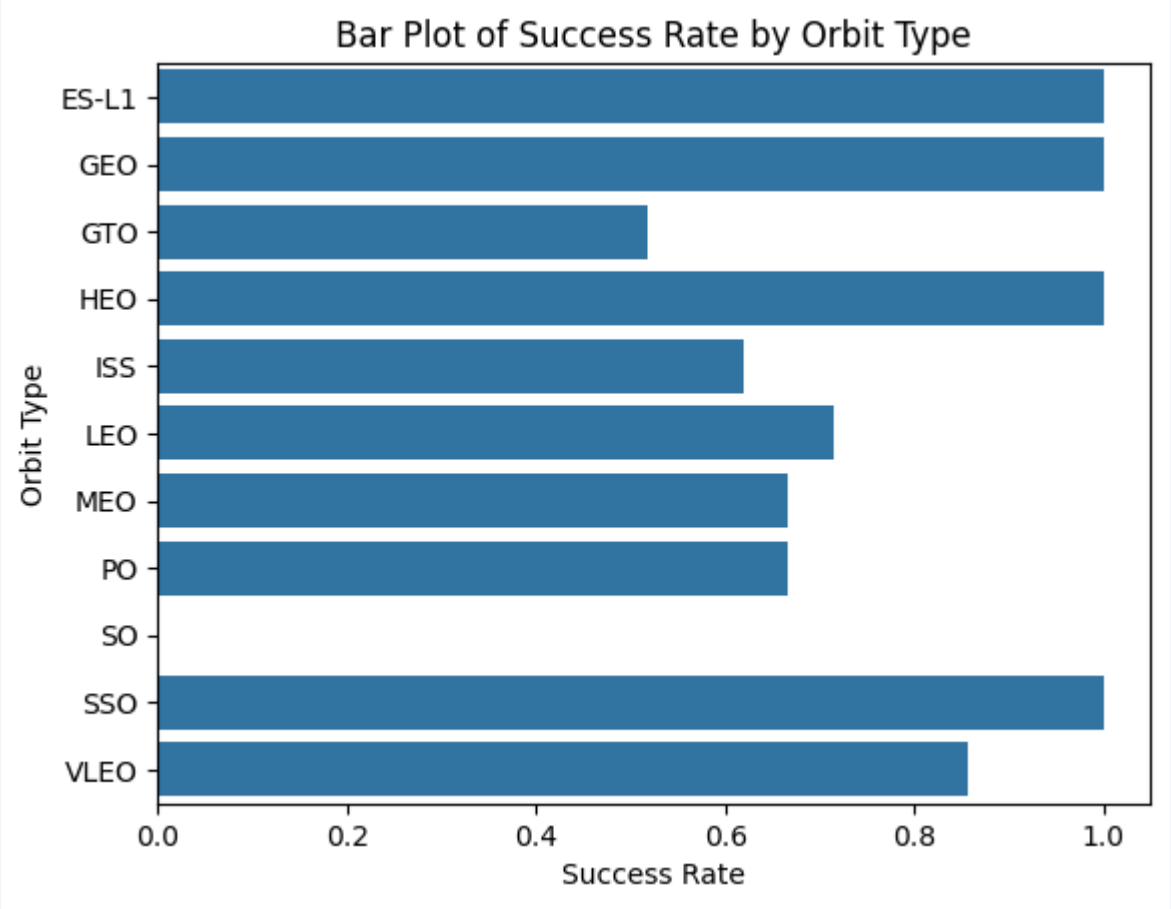
Payload vs. Launch Site



- This plot shows the payload mass by launch site
  - The class is a 0 for failure and a 1 for success
- Insights:
  - Some launch sites handle more varied amounts of mass
  - High payload missions tend to be at specific sites
  - Success rate seems lower with more mass
  - Suggests a possible interaction between payload mass, launch site, and success

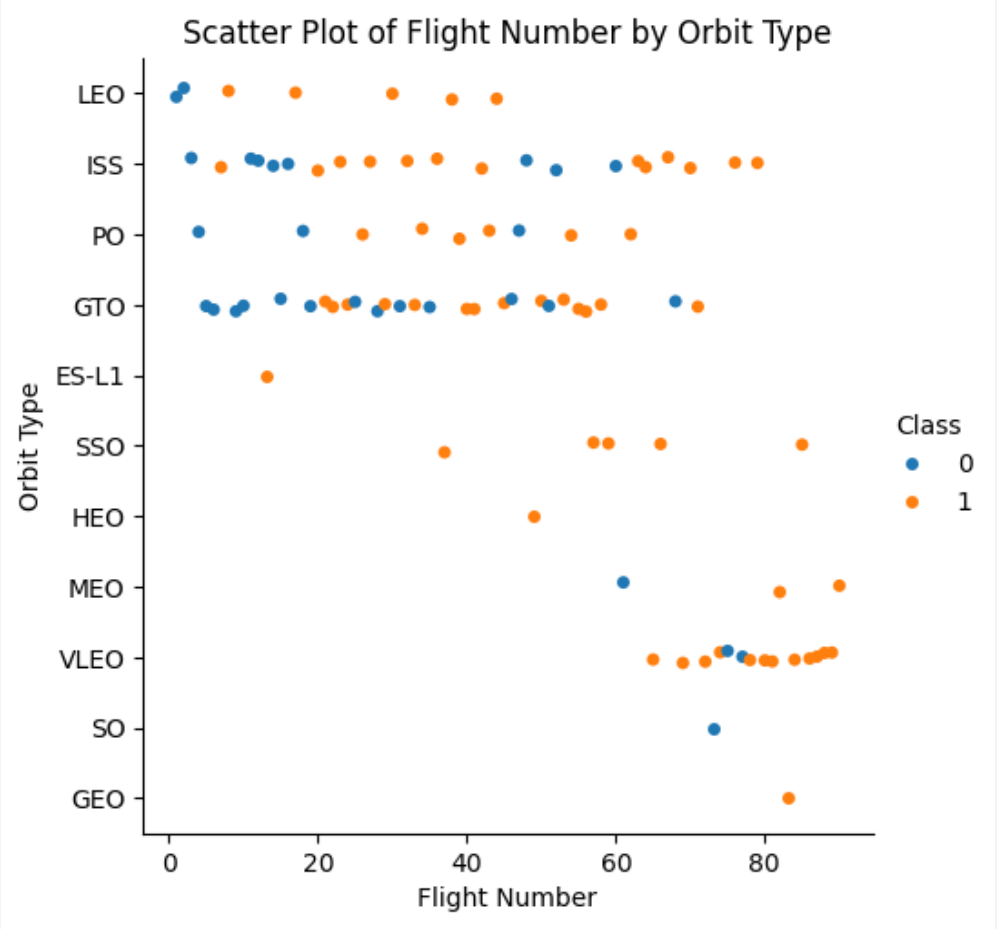
Success Rate vs. Orbit Type

- This bar chart shows the success rate of each orbit type
- Insights:
  - Certain mission types see more success (e.g. SSO, ES-L1)
  - More complex orbits such as GTO show lower success rates
  - Orbit type has significant impact on mission difficulty and outcome
  - Orbit type will be a key feature for predictive modeling



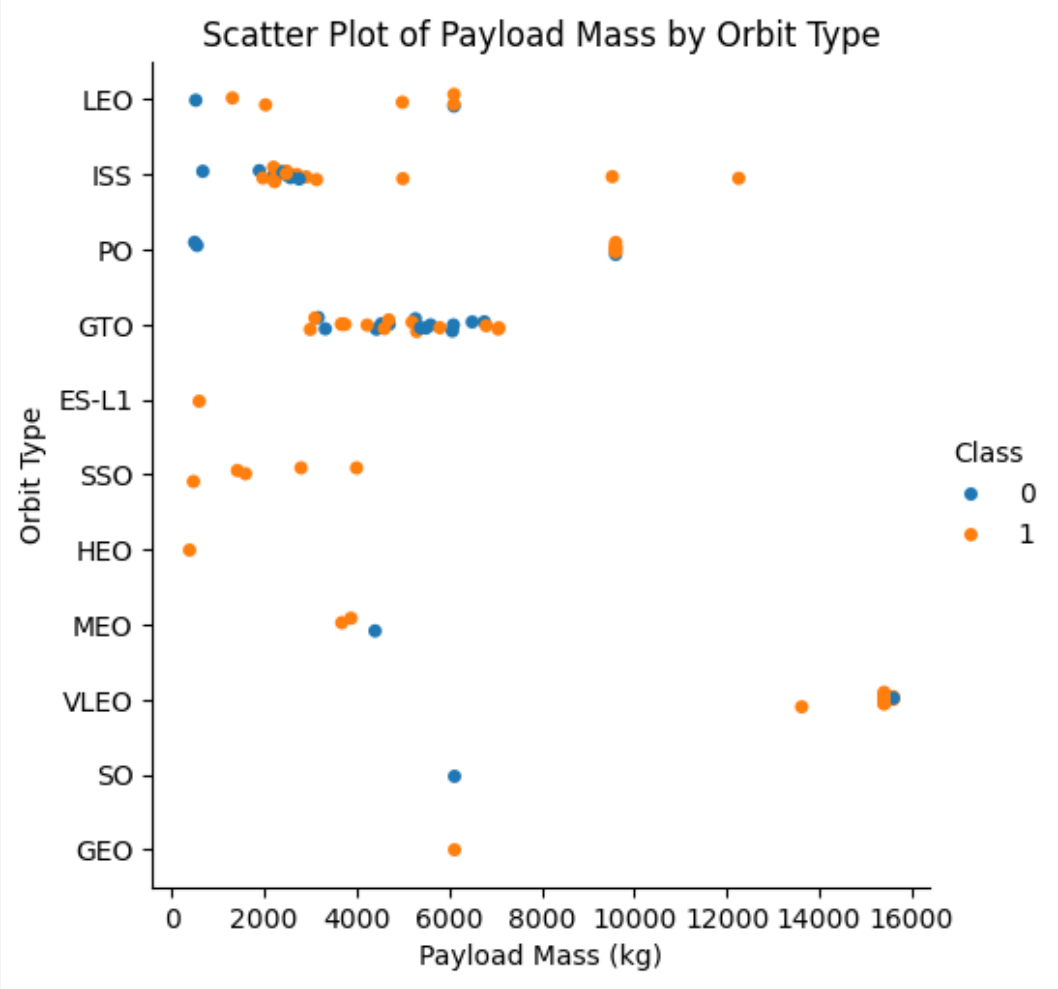
Flight Number vs. Orbit Type

- This scatter plot shows the flight number by orbit type
  - Chart shows how orbit usage changes over time
  - Shows relationship between complexity and experience (more flights => more difficult orbits)
- Insights:
  - Early missions mostly in simple orbits with some in complex ones
  - Over time, more complex orbits appear more frequently
  - Indicates a learning effect and technological improvement



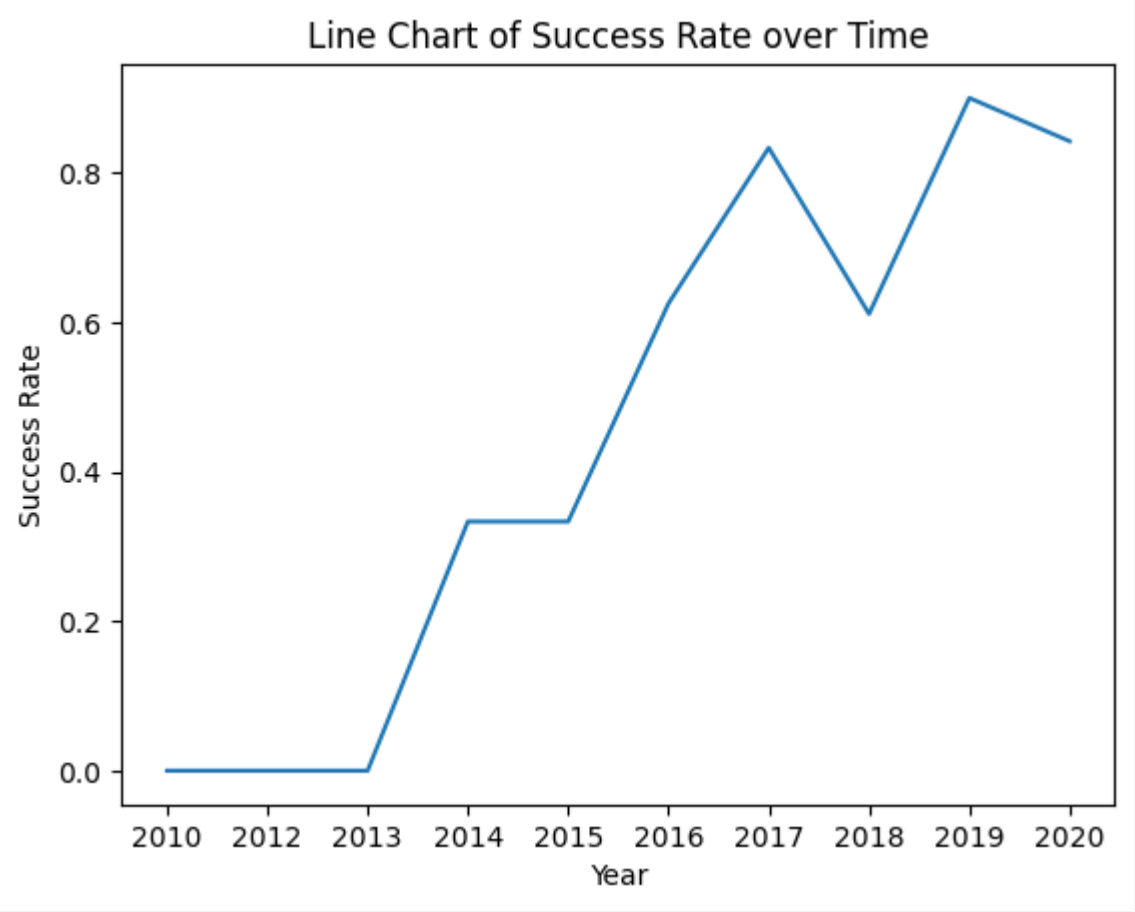
Payload vs. Orbit Type

- This scatterplot shows the payload mass by orbit
  - Shows distribution of mass by orbit type
  - Shows relationship between mission complexity and payload
- Insights
  - Higher payload missions are less common overall, and primarily only appear in VLEO
  - Lower payload missions are concentrated in LEO, ISS, and GTO orbits
  - Success rates vary depending on both payload mass and orbit
  - Shows interaction between mission complexity and performance
  - Payload and orbit together are strong indicators of success



Launch Success Yearly Trend

- This chart shows launch success rate over time
- Insights
  - Success rate significantly improving over time
  - Indicates strong learning curve and technological improvement
  - Confirms SpaceX has increasing reliability
  - Suggests experience (flight number) is a success factor



All Launch Site Names

- Unique launch sites from SQL:
  - CCAFS LC-40
  - VAFB SLC-4E
  - KSC LC-39A
  - CCAFS SLC-40
- This simple query established the set of locations for launches
- The names also provide context for site-based EDA and modeling

Launch Site Names Begin with 'CCA'

- First 5 records where launch sites begin with `CCA`:
  - ('2010-06-04', '18:45:00', 'F9 v1.0 B0003', 'CCAFS LC-40', 'Dragon Spacecraft Qualification Unit', 0, 'LEO', 'SpaceX', 'Success', 'Failure (parachute)')
  - ('2010-12-08', '15:43:00', 'F9 v1.0 B0004', 'CCAFS LC-40', 'Dragon demo flight C1, two CubeSats, barrel of Brouere cheese', 0, 'LEO (ISS)', 'NASA (COTS) NRO', 'Success', 'Failure (parachute)')
  - ('2012-05-22', '7:44:00', 'F9 v1.0 B0005', 'CCAFS LC-40', 'Dragon demo flight C2', 525, 'LEO (ISS)', 'NASA (COTS)', 'Success', 'No attempt')
  - ('2012-10-08', '0:35:00', 'F9 v1.0 B0006', 'CCAFS LC-40', 'SpaceX CRS-1', 500, 'LEO (ISS)', 'NASA (CRS)', 'Success', 'No attempt')
  - ('2013-03-01', '15:10:00', 'F9 v1.0 B0007', 'CCAFS LC-40', 'SpaceX CRS-2', 677, 'LEO (ISS)', 'NASA (CRS)', 'Success', 'No attempt')
- This query filters launch sites whose names begin with “CCA”
- The results show launches conducted at Cape Canaveral Air Force Station (CCAFS)
- These sites are among the most frequently used launch locations

Total Payload Mass

- Calculated total payload carried by boosters from NASA:
  - (45596,)
- This query calculates the total payload mass (kg) carried for NASA missions
- This provides a measure of overall mission scale and operational capacity

Average Payload Mass by F9 v1.1

- Calculated average payload mass carried by booster version F9 v1.1:
  - (2928.4,)
- This number shows us the average mass carried by a particular rocket booster
- The result indicates the typical payload capacity of this version
  - This can help compare performance across rocket versions

First Successful Ground Landing Date

- Date of the first successful landing outcome:
  - ('2010-06-04',)
- This query identifies the first successful ground pad landing
- This date marks a key milestone in SpaceX's reusable rocket program

Successful Drone Ship Landing with Payload between 4000 and 6000

- Names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000
  - ('F9 FT B1022',)
  - ('F9 FT B1026',)
  - ('F9 FT B1021.2',)
  - ('F9 FT B1031.2’,)
- This query result highlights missions with moderate payloads
- This helps analyze success under controlled payload mass

Total Number of Successful and Failure Mission Outcomes

- Total number of successful and failure mission outcomes:
  - ('Failure (in flight)', 1)
  - ('Success', 98)
  - ('Success ', 1)
  - ('Success (payload status unclear)', 1)
- The results show a significantly higher number of successful launches compared to failures
- Minor inconsistencies in labeling (e.g., “Success ” vs “Success”) indicate data quality issues
- Categories could be consolidated during preprocessing for more accurate analysis

Boosters Carried Maximum Payload

- Booster versions which have carried the maximum payload mass:
  - ('F9 B5 B1048.4',)
  - ('F9 B5 B1049.4',)
  - ('F9 B5 B1051.3',)
  - ('F9 B5 B1056.4',)
  - ('F9 B5 B1048.5',)
  - ('F9 B5 B1051.4',)
  - ('F9 B5 B1049.5',)
  - ('F9 B5 B1060.2 ',)
  - ('F9 B5 B1058.3 ',)
  - ('F9 B5 B1051.6',)
  - ('F9 B5 B1060.3',)
  - ('F9 B5 B1049.7 ',)

- These results highlight the highest-capacity missions in the dataset
- This helps understand the upper-performance limits of the SpaceX launches

2015 Launch Records

- List of the failed landing outcomes in drone ship, their booster versions, and launch site names for in year 2015
  - ('01', 'Failure (drone ship)', 'F9 v1.1 B1012', 'CCAFS LC-40')
  - ('04', 'Failure (drone ship)', 'F9 v1.1 B1015', 'CCAFS LC-40')
- These results reflect some early challenges with drone ship landings
- This provides insight into the developmental phase of reusable landings

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Ranks of the count of landing outcomes between the date 2010-06-04 and 2017-03-20, in descending order
  - ('No attempt', 10)
  - ('Success (drone ship)', 5)
  - ('Failure (drone ship)', 5)
  - ('Success (ground pad)', 3)
  - ('Controlled (ocean)', 3)
  - ('Uncontrolled (ocean)', 2)
  - ('Failure (parachute)', 2)
  - ('Precluded (drone ship)', 1)

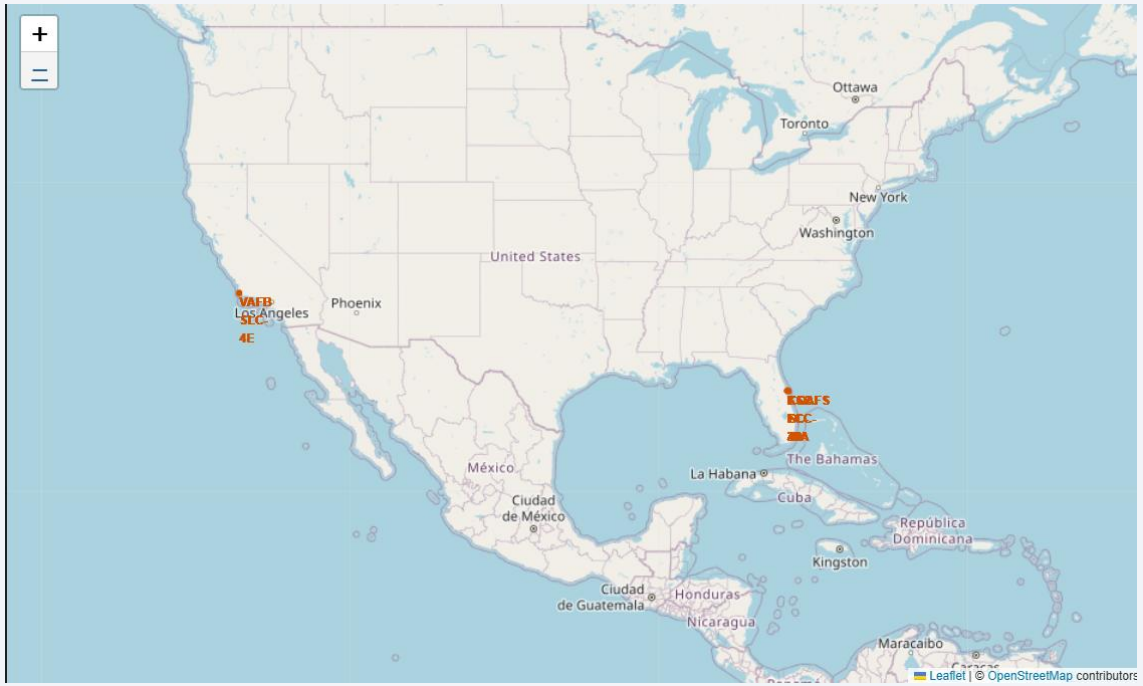
- These results show which outcomes occur most often
- This also helps identify landing and performance reliability

Section 3

Launch Sites Proximities Analysis

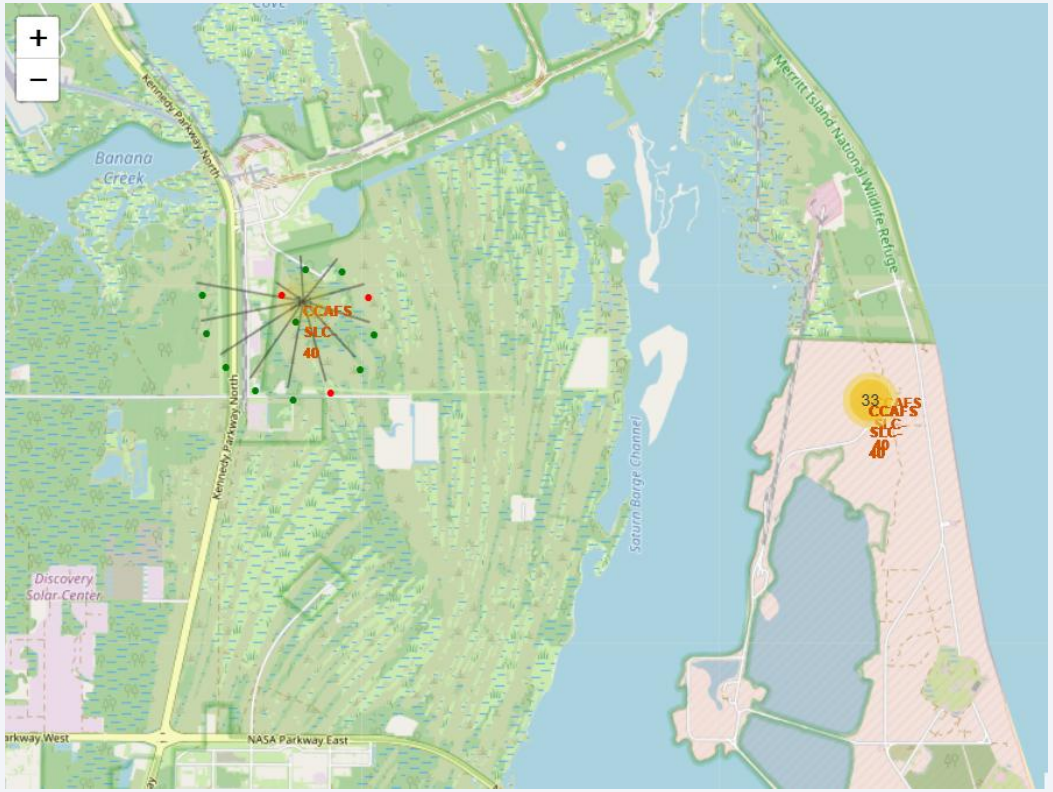
Launch Site Locations – NA

- This map displays all SpaceX locations using markers
- Launch sites are along coastlines in lower latitudes relative to the rest of the US
- Coastal placement supports safe launches over water



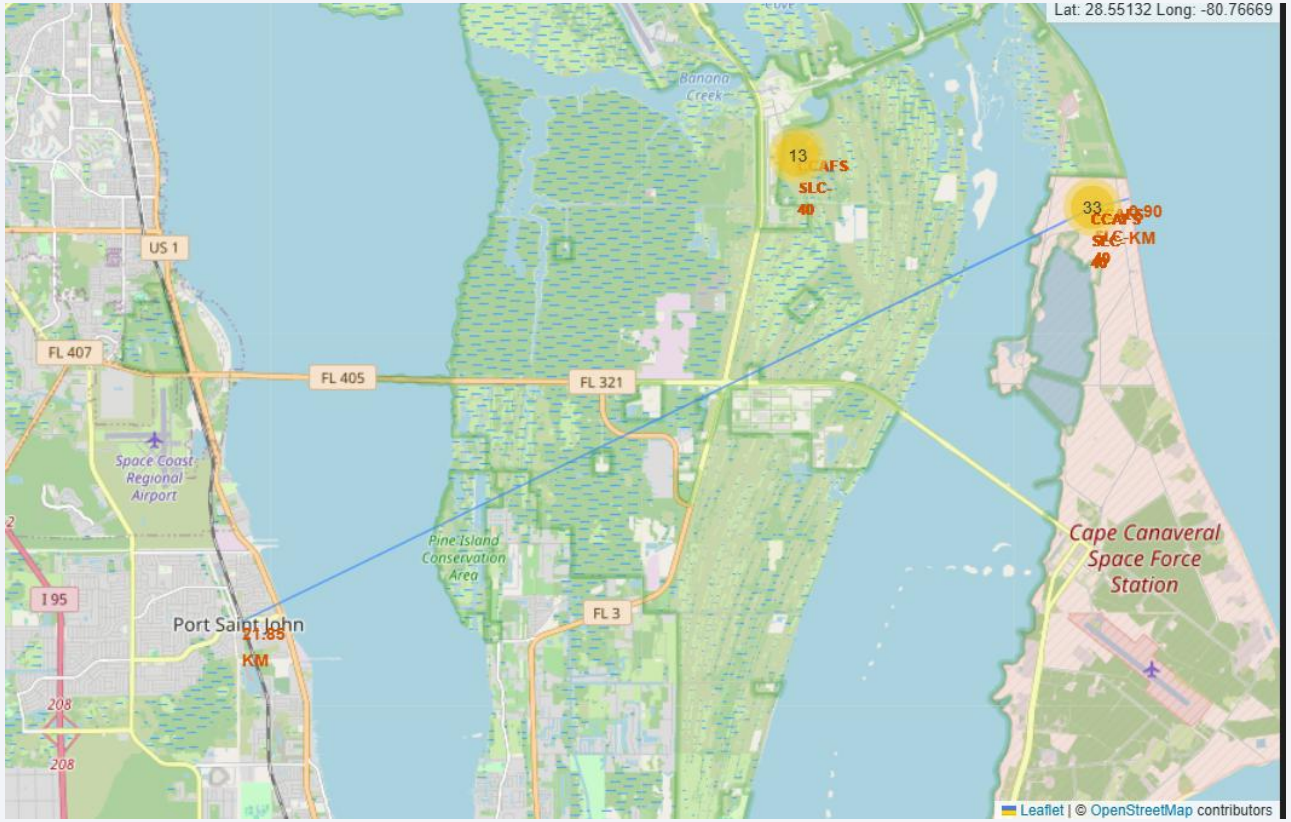
Launch Site Outcomes – Cape Canaveral

- This map uses color coded markers to represent successful and failed launches
- This map is zoomed into Cape Canaveral as there are two sites in proximity
- Launch failures are uncommon and scattered across locations
- Indicates high reliability across sites



Launch Site Proximity Analysis

- This map shows distance between the CCAFS site and nearby features, such as Port Saint John and the coastline
- Launch sites are closer to coastlines, minimizing risk
- Proximity to transportation infrastructure supports efficient operations

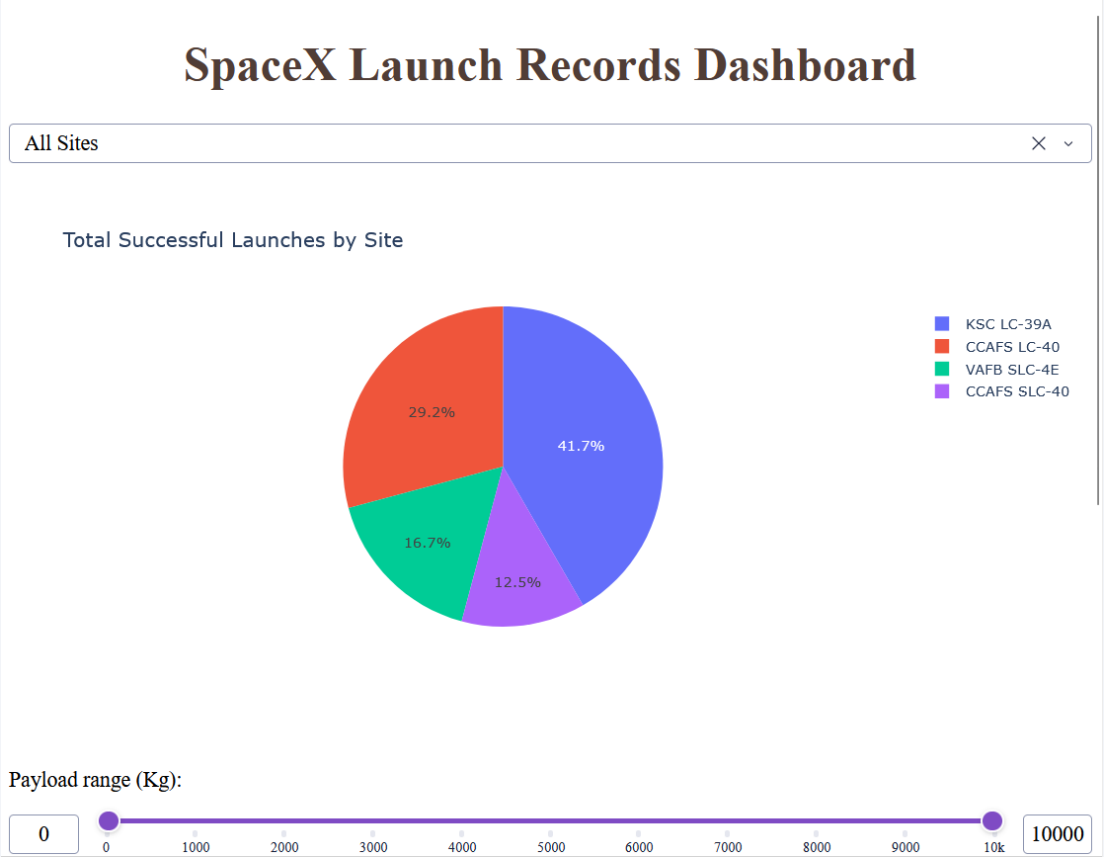


Section 4

Build a Dashboard with Plotly Dash

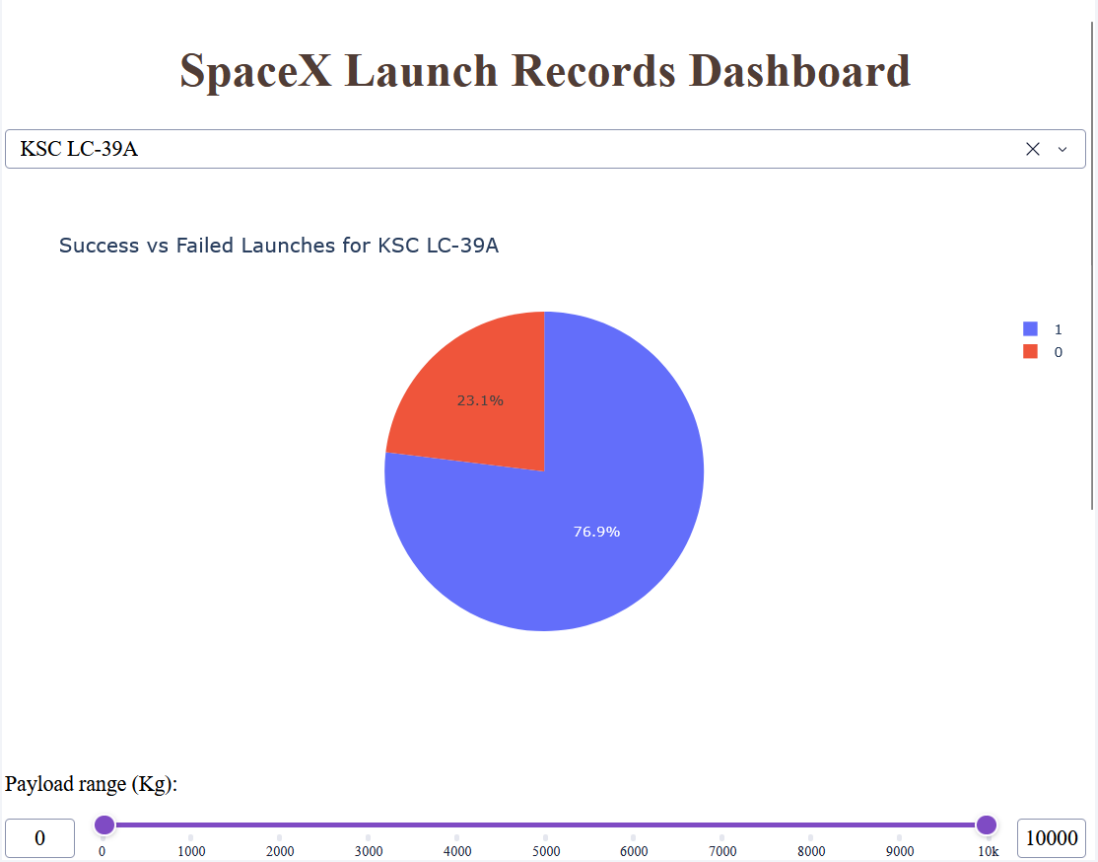
Launch Success Distribution – All Sites

- The pie chart shows the distribution of successful launches across all sites
- The largest percentage of launches come from site KSC LC-39A
- This chart provides a high-level overview of mission outcomes



Launch Success Rate – KSC LC-39A

- This pie chart shows the success vs failure distribution for KSC LC-39A
  - This site displayed the highest success ratio
- The selected site demonstrates a high success ratio compared to failures
- Highlights differences in performance across individual launch locations



Payload Mass vs Launch Outcome

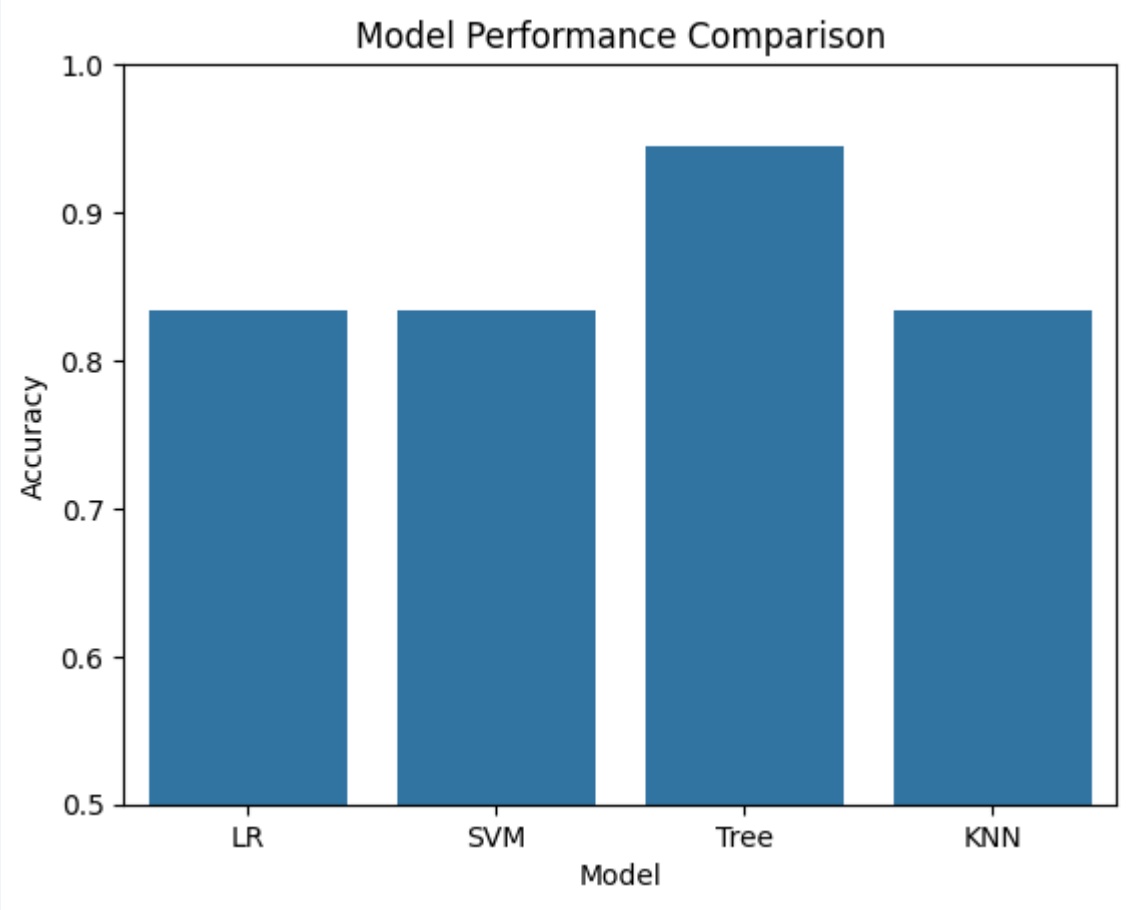
- This scatter plot shows the relationship between payload mass and launch success
- The range slider allows filtering by payload mass for dynamic analysis
- Certain payload ranges show higher concentrations of successful launches
- Heavier payloads are associated with increased mission complexity and varied outcome

Section 5

Predictive Analysis (Classification)

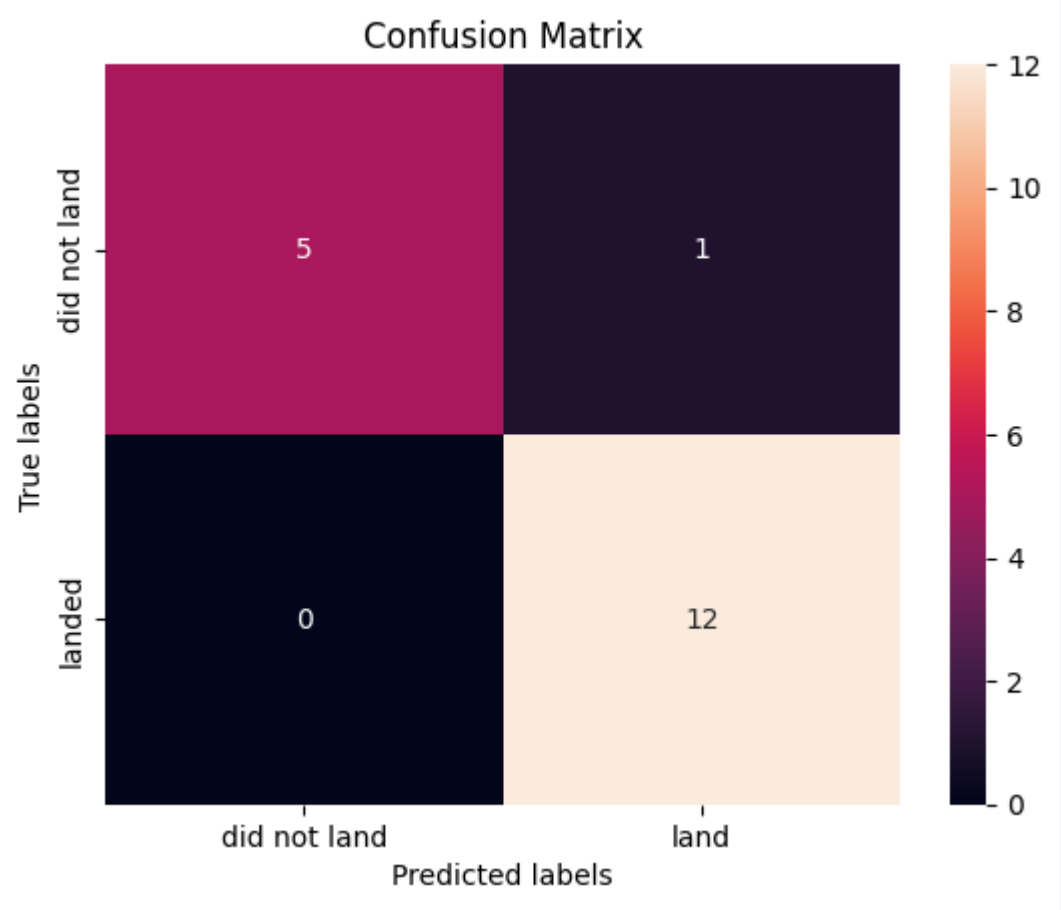
Classification Accuracy

- The bar chart compares the accuracy of multiple classification models
- Models included are Logistic Regression (LR), Support Vector Machine (SVM), Decision Tree (Tree), and K-Nearest Neighbors (KNN)
- The Decision Tree achieved the highest accuracy and was selected as the final mode



Confusion Matrix

- This confusion matrix shows the classification performance of the Decision Tree
- The model correctly predicts the majority of successful and failed launches
- High values along the diagonal indicate strong predictive performance
  - True positives and true negatives are dominant



Conclusions

- Launch success rate has improved significantly over time
- Payload mass, orbit type, and booster version are key predictors
- Interactive analysis (map and dashboard) enhance context and understanding of launch patterns
- Machine learning models can effectively predict launch success
- Data-driven insights can support future mission planning and optimization
- SpaceX's success is strongly driven by experience, mission parameters, and technological advancement

Appendix

- GitHub full project repo:
  - [Repository Link]
- Sources include:
  - Wikipedia contributors. (n.d.). *List of Falcon 9 and Falcon Heavy launches*. Wikipedia. Retrieved from https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches
  - SpaceX. (n.d.). *SpaceX API*. Retrieved from https://api.spacexdata.com/
- Example code snippets (bar plot, svm model, sql query):

```python
orbit_success = df.groupby("Orbit")["Class"].mean().reset_index()

sns.barplot(y="Orbit", x="Class", data=orbit_success)
plt.title("Bar Plot of Success Rate by Orbit Type")
plt.xlabel("Success Rate",fontsize=10)
plt.ylabel("Orbit Type",fontsize=10)
plt.show()
```

```python
parameters = {'kernel':('linear', 'rbf','poly','rbf', 'sigmoid'),
              'C': np.logspace(-3, 3, 5),
              'gamma':np.logspace(-3, 3, 5)}
svm = SVC()
```

```python
svm_cv = GridSearchCV(estimator=svm, param_grid=parameters, cv=10)
svm_cv.fit(X_train, Y_train)
```

```python
query = """
SELECT Booster_Version
FROM SPACEXTABLE
WHERE PAYLOAD_MASS__KG_ = (
    SELECT MAX(PAYLOAD_MASS__KG_)
    FROM SPACEXTABLE)
"""
cur.execute(query)
rows = cur.fetchall()
for row in rows:
    print(row)
```

```
('F9 B5 B1048.4',)
('F9 B5 B1049.4',)
```

Thank you!